

1 ECR Data-Driven Targeting

1.1 Plant, surface of section and the controlled map

The Lorenz system is written in full, since the second equation is what defines the surface of section used below,

$$\dot{x} = \sigma(y - x), \quad (1)$$

$$\dot{y} = \rho x - y - xz, \quad (2)$$

$$\dot{z} = xy - \beta z, \quad (3)$$

with state $\mathbf{u} = [x \ y \ z]^\top \in \mathbb{R}^3$ and control parameter vector

$$\mathbf{p} = [\sigma \ \rho \ \beta]^\top \in \mathbb{R}^r, \quad r = 3. \quad (4)$$

Let $\varphi_t(\mathbf{u}; \mathbf{p})$ denote the flow of (1)–(3). The surface of section is the half-plane

$$\Sigma = \{\mathbf{u} \in \mathbb{R}^3 : y = y_{\text{PSS}}, \ \dot{y} < 0\}, \quad y_{\text{PSS}} = \sqrt{\beta_{\text{nom}}(\rho_{\text{nom}} - 1)} = 8.4853, \quad (5)$$

i.e. the y -coordinate of the non-trivial equilibria $C^\pm$. The sign condition selects one of the two transversal branches; by (2) transversality on Σ requires

$$\dot{y}|_\Sigma = \rho x - y_{\text{PSS}} - xz \neq 0, \quad (6)$$

which holds on the whole attracting set of interest. The branch $\dot{y} < 0$ is the one that reproduces the target and the state RMS quoted in the source paper; on the branch $\dot{y} > 0$ the equilibrium C^+ itself lies in Σ and the fixed-point problem below degenerates onto it.

With the first return time

$$\tau(\mathbf{u}; \mathbf{p}) = \min\{t > 0 : \varphi_t(\mathbf{u}; \mathbf{p}) \in \Sigma\}, \quad (7)$$

the Poincaré map is $\mathbf{u} \mapsto \varphi_\tau(\mathbf{u}; \mathbf{p})$. Because $y \equiv y_{\text{PSS}}$ on Σ , the section state is fully described by two coordinates, and the *observed state* of the controller is taken as

$$\mathbf{z}_n = [x(t_n) \ z(t_n)]^\top \in \mathbb{R}^N, \quad N = 2, \quad (8)$$

where t_n is the instant of the n -th crossing. The controlled dynamics therefore reduce to the discrete map

$$\mathbf{z}_{n+1} = \mathbf{G}(\mathbf{z}_n, \mathbf{p}_n), \quad (9)$$

in which the parameter vector is held constant over the whole flight $t \in [t_n, t_{n+1})$ and may be changed only at crossings. Equation (9) is the object the ECR method learns; no closed form for $\mathbf{G}$ is ever constructed.

Delay-coordinate variant. When only a scalar measurement $s(t) = g(\mathbf{u}(t))$ is available, the observed state is the delay embedding evaluated at the crossings,

$$\mathbf{z}_n = [s(t_n) \ s(t_n - T) \ \cdots \ s(t_n - (N - 1)T)]^\top, \quad (10)$$

with delay T and embedding dimension N . By Takens' theorem (10) is generically diffeomorphic to the section state, so (9) still holds — with the caveat discussed in Remark 3.

1.2 Admissible parameter set

Each parameter may be perturbed about its nominal value within a prescribed range,

$$\Pi = \left\{ \mathbf{p} \in \mathbb{R}^r : |p^i - p_{\text{nom}}^i| < \delta p_{\text{max}}^i, \quad i = 1, \dots, r \right\}, \quad (11)$$

with, for the Lorenz system, $\mathbf{p}_{\text{nom}} = [10 \ 28 \ 8/3]^\top$ and $\delta \mathbf{p}_{\text{max}} = [0.30 \ 0.84 \ 0.08]^\top$, i.e. exactly 3% of each nominal value. It is convenient to work with the normalised perturbation

$$\Delta \mathbf{p}_n = (\mathbf{p}_n - \mathbf{p}_{\text{nom}}) \oslash \delta \mathbf{p}_{\text{max}}, \quad \mathbf{p}_n \in \Pi \iff \|\Delta \mathbf{p}_n\|_\infty \leq 1, \quad (12)$$

where $\oslash$ is the elementwise quotient. Every learned model below outputs $\Delta \mathbf{p}$, so admissibility is a box test on the model output.

1.3 Target orbit and its local geometry

The target is a fixed point of the controlled map at nominal parameters,

$$\mathbf{z}^* = \mathbf{G}(\mathbf{z}^*, \mathbf{p}_{\text{nom}}), \quad (13)$$

which corresponds to a period-1 unstable periodic orbit of the flow (not to an equilibrium: the equilibria of (1)–(3) do not lie on the branch $\dot{y} < 0$ of Σ). It is located by Newton iteration on the residual $\mathbf{F}(\mathbf{z}) = \mathbf{G}(\mathbf{z}, \mathbf{p}_{\text{nom}}) - \mathbf{z}$,

$$\mathbf{z}^{(k+1)} = \mathbf{z}^{(k)} - (\mathbf{J} - \mathbf{I})^{-1} \mathbf{F}(\mathbf{z}^{(k)}), \quad \mathbf{J} = \left. \frac{\partial \mathbf{G}}{\partial \mathbf{z}} \right|_{(\mathbf{z}^*, \mathbf{p}_{\text{nom}})}, \quad (14)$$

the Jacobian being evaluated by central differences of the numerically integrated map. This yields

$$\mathbf{z}^* = \begin{bmatrix} 14.2522 \\ 39.7867 \end{bmatrix}, \quad \tau^* = 1.5587, \quad \mathbf{J} = \begin{bmatrix} 0.9594 & 1.7943 \\ 2.0070 & 3.7536 \end{bmatrix}, \quad (15)$$

with eigenvalues

$$\lambda_1 = 4.7129, \quad \lambda_2 \approx 1.3 \times 10^{-9}, \quad (16)$$

so the target is a saddle whose stable direction is contracted so violently that the return map is effectively one-dimensional — the reason why control regions on this section appear as thin filaments along the attractor.

The actuation available in one step is described by the sensitivity matrix

$$\mathbf{B} = \left. \frac{\partial \mathbf{G}}{\partial \mathbf{p}} \right|_{(\mathbf{z}^*, \mathbf{p}_{\text{nom}})} = \begin{bmatrix} 0.1739 & -2.3023 & -4.0222 \\ 0.2353 & -3.9651 & -10.3387 \end{bmatrix}, \quad \text{rank } \mathbf{B} = 2, \quad (17)$$

whose singular values are 11.99 and 0.659. Since $\text{rank } \mathbf{B} = N$, the one-step image can be displaced in any direction of the section plane; the reachable displacement under the full parameter budget,

$$|\mathbf{B}| \delta \mathbf{p}_{\text{max}} = [2.31 \ 4.23]^\top, \quad (18)$$

is an order of magnitude larger than the target radius $\delta = 0.30$ used below, which is what makes single-step targeting into T_0 possible at all. Note that σ is by far the weakest actuator: alone it moves the image by less than δ .

1.4 Control regions

Let $\delta > 0$ be the target radius and define the δ -ball $\mathcal{B}_\delta = \{\mathbf{z} : \|\mathbf{z} - \mathbf{z}^*\| < \delta\}$.

Definition 1 (Target region, T_0).

$$T_0 = S_0 = \left\{ \mathbf{z} \in \mathcal{B}_\delta : \exists \mathbf{p} \in \Pi \text{ such that } \mathbf{G}(\mathbf{z}, \mathbf{p}) \in \mathcal{B}_\delta \right\}. \quad (19)$$

T_0 is thus the set of states that are both close to the target and can be kept close by an admissible perturbation; it is the region in which the local (fine) controller operates.

Definition 2 (Extended control regions, T_i). For $i = 1, 2, \dots, K$,

$$T_i = \left\{ \mathbf{z} \notin \bigcup_{j < i} T_j : \exists \mathbf{p} \in \Pi \text{ such that } \mathbf{G}(\mathbf{z}, \mathbf{p}) \in T_{i-1} \right\}. \quad (20)$$

Equivalently, T_i is the set of states that are one admissible step from T_{i-1} and not already members of a lower region. The regions are disjoint by construction, and

$$\mathbf{z}_n \in T_i \implies \exists \{\mathbf{p}_n, \dots, \mathbf{p}_{n+i-1}\} \subset \Pi : \mathbf{G}(\dots \mathbf{G}(\mathbf{z}_n, \mathbf{p}_n) \dots, \mathbf{p}_{n+i-1}) \in T_0, \quad (21)$$

i.e. membership of T_i certifies capture of the target region in at most i steps. The union $\bigcup_{i=0}^K T_i$ is the activation region of the controller: outside it the nominal parameters are applied.

Remark 1. Definitions 1 and 2 are stated with $\mathbf{G}$, but they are existence statements about admissible perturbations, not formulae. Section 1.5 replaces each existential quantifier by evidence drawn from measured transitions, which is what makes the method model-free.

1.5 Data model and the empirical construction of the regions

The experiment is a random-perturbation sweep: the plant is run from many initial conditions on the attractor while, at every crossing, a parameter vector is drawn independently and uniformly from Π . The resulting data set is a collection of transition triples

$$\mathcal{D} = \{(\mathbf{z}_n, \mathbf{p}_n, \mathbf{z}_{n+1})\}_{n=1}^M, \quad \mathbf{p}_n \sim \mathcal{U}(\Pi), \quad \mathbf{z}_{n+1} = \mathbf{G}(\mathbf{z}_n, \mathbf{p}_n). \quad (22)$$

Region membership is then decided on $\mathcal{D}$ alone. Level 0 uses Definition 1 directly, the applied $\mathbf{p}_n$ serving as the witness of the existential quantifier:

$$\mathcal{I}_0 = \{n : \|\mathbf{z}_n - \mathbf{z}^*\| < \delta \wedge \|\mathbf{z}_{n+1} - \mathbf{z}^*\| < \delta\}. \quad (23)$$

For $i \geq 1$ the test requires membership of the *successor* in T_{i-1} , a set that is itself known only through data. This is resolved by using the already-trained level- $(i-1)$ model as the membership oracle. Writing $\mathcal{M}_{i-1} : \mathbb{R}^N \rightarrow \{0, 1\}$ for that oracle (defined in (33) below),

$$\mathcal{I}_i = \left\{ n : \mathcal{M}_{i-1}(\mathbf{z}_{n+1}) = 1 \wedge \mathcal{M}_j(\mathbf{z}_n) = 0 \ \forall j < i \right\}. \quad (24)$$

The training pairs of region i are $\{(\mathbf{z}_n, \Delta \mathbf{p}_n)\}_{n \in \mathcal{I}_i}$: the state, and the normalised perturbation that was observed to move it into the next lower region. Construction proceeds bottom-up, $i = 0, 1, \dots, K$, each level supplying the oracle for the next.

1.6 Clustering of a control region (ECR-III)

ECR-I models the whole activation region with a single network and ECR-II one network per region T_i . ECR-III first partitions each region into clusters, on the grounds that data belonging to the higher regions is scattered and a single interpolant across disjoint patches is meaningless.

Let $\mathcal{Z}_i = \{\mathbf{z}_n\}_{n \in \mathcal{I}_i}$. Choose a radius r_i and define on $\mathcal{Z}_i$ the relation

$$\mathbf{z} \sim \mathbf{z}' \iff \|\mathbf{z} - \mathbf{z}'\| \leq r_i. \quad (25)$$

The clusters $C_{i1}, \dots, C_{in_i}$ are the equivalence classes of the transitive closure of (25); equivalently, two points lie in the same cluster iff they are joined by a chain of points spaced by at most r_i (single-linkage at threshold r_i). The radius is selected from the empirical distribution of pairwise distances: with

$$h_i(\cdot) = \text{histogram of } \{\|\mathbf{z} - \mathbf{z}'\| : \mathbf{z}, \mathbf{z}' \in \mathcal{Z}_i, \mathbf{z} \neq \mathbf{z}'\}, \quad (26)$$

r_i is taken at the first local minimum of h_i following its first local maximum — the gap that separates intra-cluster from inter-cluster distances.

Each cluster is summarised by its sample mean and covariance,

$$\boldsymbol{\mu}_{ij} = \frac{1}{|C_{ij}|} \sum_{\mathbf{z} \in C_{ij}} \mathbf{z}, \quad \boldsymbol{\Sigma}_{ij} = \frac{1}{|C_{ij}| - 1} \sum_{\mathbf{z} \in C_{ij}} (\mathbf{z} - \boldsymbol{\mu}_{ij})(\mathbf{z} - \boldsymbol{\mu}_{ij})^\top + \varepsilon \frac{\text{tr } \boldsymbol{\Sigma}_{ij}}{N} \mathbf{I}, \quad (27)$$

the last term being a ridge that keeps $\boldsymbol{\Sigma}_{ij}$ invertible for the thin, nearly one-dimensional clusters produced by a strongly contracting map. The region of a current state is then decided by the *normalised Mahalanobis distance*

$$\text{NMD}_{ij}^2(\mathbf{z}_n) = (\mathbf{z}_n - \boldsymbol{\mu}_{ij})^\top \underline{\mathbf{N}}_{ij} (\mathbf{z}_n - \boldsymbol{\mu}_{ij}), \quad (28)$$

where $\underline{\mathbf{N}}_{ij}$ is the normalised covariance matrix. The normalisation is not fixed by the source paper; the implementation supports

$$\underline{\mathbf{N}}_{ij} = \begin{cases} (\boldsymbol{\Sigma}_{ij} / \det(\boldsymbol{\Sigma}_{ij})^{1/N})^{-1} & \text{unit determinant (default),} \\ \boldsymbol{\Sigma}_{ij}^{-1} & \text{plain Mahalanobis,} \\ \text{diag}(1/\Sigma_{ij,11}, \dots, 1/\Sigma_{ij,NN}) & \text{normalised variances,} \\ (\boldsymbol{\Sigma}_{ij} N / \text{tr } \boldsymbol{\Sigma}_{ij})^{-1} & \text{unit trace.} \end{cases} \quad (29)$$

The first and fourth remove the *size* of a cluster while retaining its shape, so that clusters of different populations compete on equal terms; for an isotropic cluster the first reduces to the Euclidean distance.

1.7 Local models

Each cluster carries a Gaussian radial-basis-function network mapping a state to a normalised perturbation. With standardised inputs $\tilde{\mathbf{z}} = (\mathbf{z} - \bar{\mathbf{z}}) \oslash \mathbf{s}$, centres $\mathbf{c}_k$ and widths s_k ,

$$\phi_k(\mathbf{z}) = \exp\left(-\frac{\|\tilde{\mathbf{z}} - \mathbf{c}_k\|^2}{2s_k^2}\right), \quad k = 1, \dots, n_c, \quad (30)$$

$$\widehat{\Delta \mathbf{p}} = \mathbf{W}^\top [\phi_1(\mathbf{z}) \cdots \phi_{n_c}(\mathbf{z}) \ 1]^\top, \quad \hat{\mathbf{p}} = \mathbf{p}_{\text{nom}} + \widehat{\Delta \mathbf{p}} \odot \delta \mathbf{p}_{\text{max}}. \quad (31)$$

Centres are placed by k -means on the cluster data, widths from the mean distance to the nearest centres, and the output weights follow in closed form from regularised least squares,

$$\mathbf{W} = (\Phi^\top \Phi + \lambda \mathbf{I})^{-1} \Phi^\top \mathbf{Y}, \quad \Phi_{nk} = \phi_k(\mathbf{z}_n), \quad \mathbf{Y}_{n:} = \Delta \mathbf{p}_n^\top, \quad (32)$$

which is why ECR training costs seconds rather than the iterative effort of a back-propagation network. Because the target of the regression is a normalised perturbation, admissibility is the box test of (12), and the membership oracle used in (24) is simply

$$\mathcal{M}_i(\mathbf{z}) = \begin{cases} 1, & \text{if } \|\widehat{\Delta \mathbf{p}}^{(ij^*)}(\mathbf{z})\|_\infty \leq 1 \text{ for the cluster } j^* = \arg \min_j \text{NMD}_{ij}(\mathbf{z}), \\ 0, & \text{otherwise,} \end{cases} \quad (33)$$

optionally restricted to clusters for which $\text{NMD}_{ij}(\mathbf{z}) \leq \gamma q_{95}(C_{ij})$, where q_{95} is the 95th percentile of the NMD of the cluster's own training data and γ a gate factor. The gate exists because (28) always returns a nearest cluster, including for states nowhere near any recorded data, for which the network extrapolates.

1.8 On-line control law

At each crossing the controller measures $\mathbf{z}_n$, identifies a region, and applies the perturbation that region prescribes. The three variants differ only in how the region is identified.

ECR-I. One network is trained over the whole activation region and evaluated directly; the current region is never identified explicitly.

ECR-II. The state is offered to the networks in ascending region order and the first admissible answer is accepted:

$$i^* = \min\{i \in \{0, \dots, K\} : \|\widehat{\Delta \mathbf{p}}^{(i)}(\mathbf{z}_n)\|_\infty \leq 1\}. \quad (34)$$

ECR-III. The region follows analytically from (28), the search running over the clusters of *all* regions at once:

$$(i^*, j^*) = \arg \min_{i,j} \text{NMD}_{ij}(\mathbf{z}_n). \quad (35)$$

In every variant the applied parameter is

$$\mathbf{p}_n = \begin{cases} \mathbf{p}_{\text{nom}} + \widehat{\Delta \mathbf{p}}^{(i^*j^*)}(\mathbf{z}_n) \odot \delta \mathbf{p}_{\text{max}}, & \text{if such an } (i^*, j^*) \text{ exists,} \\ \mathbf{p}_{\text{nom}}, & \text{otherwise (no targeting),} \end{cases} \quad (36)$$

so the perturbation is admissible by construction and the plant runs at nominal parameters whenever the state is unrecognised.

1.9 Performance measures

For an initial condition $\mathbf{z}_0$ the *reaching time* is the number of map steps needed to enter the target region,

$$n_r(\mathbf{z}_0) = \min\{n \geq 0 : \|\mathbf{z}_n - \mathbf{z}^*\| < \delta\}, \quad (37)$$

and the reported figure of merit is its mean over a fixed ensemble $\{\mathbf{z}_0^{(m)}\}_{m=1}^{M_{\text{ic}}}$ of initial conditions drawn from the attractor, censored at a horizon n_{max} ,

$$\bar{n}_r = \frac{1}{M_{\text{ic}}} \sum_{m=1}^{M_{\text{ic}}} \min\{n_r(\mathbf{z}_0^{(m)}), n_{\text{max}}\}. \quad (38)$$

Two further quantities separate targeting from stabilisation: the *capture rate* $\Pr\{n_r < n_{\text{max}}\}$, and the *retention*

$$\eta = \frac{1}{n_h} \sum_{n=n_r+1}^{n_r+n_h} \mathbb{I}[\|\mathbf{z}_n - \mathbf{z}^*\| < \delta], \quad (39)$$

the fraction of the n_h steps following capture during which the local controller holds the state inside the target region. Without targeting, $\bar{n}_r$ measures the natural recurrence time of the section trajectory to $\mathcal{B}_\delta$; for the present configuration only 6% of crossings fall inside $\mathcal{B}_\delta$, and $\bar{n}_r \approx 15$ steps.

1.10 Algorithm

Off-line (training).

1. Collect $\mathcal{D}$ by applying $\mathbf{p}_n \sim \mathcal{U}(\Pi)$ at every crossing, (22).
2. Label $\mathcal{I}_0$ by (23); cluster $\mathcal{Z}_0$ with radius r_0 from (26); compute $(\boldsymbol{\mu}_{0j}, \mathbf{N}_{0j})$ by (27)–(29); fit NN_{0j} by (32).
3. For $i = 1, \dots, K$: label $\mathcal{I}_i$ by (24) using the oracle (33) of level $i - 1$; cluster; fit NN_{ij} . Stop when $|\mathcal{I}_i|$ falls below a minimum count.

On-line (targeting and control).

1. Measure $\mathbf{z}_n$ at the crossing.
2. Identify the region by (34) or (35).
3. Apply $\mathbf{p}_n$ from (36) and hold it until the next crossing.

1.11 Remarks on the underdetermined elements

Remark 2 (Choices not fixed by the source formulation). *The normalisation in (29), the gate factor γ , the tie-breaking rule in (35), and the selection of the training pairs for NN_{0j} are all left open in the original description. They are not cosmetic: on the Lorenz section the size-removing normalisations reach the target in 4.3–4.4 steps with retention 0.96, whereas plain Mahalanobis or normalised-variance readings require 5.2 steps and hold the orbit only 0.61–0.69 of the time, with markedly larger run-to-run spread.*

Remark 3 (Delay coordinates). *Under (10) the components $s(t_n - kT)$, $k \geq 1$, were generated while the previous parameter vector was applied. The observed transition is therefore a function of $(\mathbf{z}_n, \mathbf{p}_n, \mathbf{p}_{n-1})$ rather than of $(\mathbf{z}_n, \mathbf{p}_n)$ alone, and (9) holds only approximately. Coarse targeting tolerates this ambiguity — the reaching time still improves substantially — but local stabilisation does not, the retention (39) falling from ≈ 0.96 in real coordinates to ≈ 0.20 . The standard remedy is to augment the observed state with the previously applied perturbation, $[\mathbf{z}_n^\top \ \Delta \mathbf{p}_{n-1}^\top]^\top$.*

Remark 4 (Deterministic deadlock). *Because (36) is a deterministic state feedback, a trajectory can be locked into a controlled periodic orbit that never enters $\mathcal{B}_\delta$, inflating (38) through censored samples rather than through slow targeting. Applying $\mathbf{p}_{\text{nom}}$ for a single step after a fixed number of unsuccessful controlled steps, or the presence of measurement noise, removes the deadlock.*